ABERDARE TECHNICAL COLLEGE

INFORMATION COMMUNICATION TECHNOLOGY

LEVEL 5 – MODULE 3

UNIT: NETWORK DESIGN AND MANAGEMENT

UNIT CODE: 0612 451 07A

PRACTICAL ASSESSMENT 3

Time: 2 Hours

Total Marks: 50

Materials

- *Desktop computer*
- *Steady internet connection*
- *Packet sniffer tool(Wireshark)*

INSTRUCTIONS

1. Read all instructions carefully before starting the assessment.
2. Attempt all tasks.
3. Observe all laboratory safety procedures when handling networking tools and equipment.
4. Use the materials and tools provided.
5. Save all screenshots and evidence of completed tasks in the folder provided by the assessor.
6. Label all screenshots appropriately.

7. Ensure all cable terminations are tested before submission.
8. Present all completed work to the assessor for verification.
9. The assessment is out of **50 marks**.

SCENARIO

You have been employed as a Network Technician in a training institution. The institution requires network cables to be prepared and tested before deployment. Additionally, the institution requires network traffic analysis using Wireshark to troubleshoot communication between network devices.

ELEMENT COVERED: INSTALL COMPUTER NETWORK

Perform the following tasks.

TASK 1: TERMINATION OF ETHERNET CABLE

Using the materials provided, prepare and terminate a straight-through Ethernet cable following the T568B standard.

Requirements

1. Measure and cut the Ethernet cable to the required length.
2. Strip the outer jacket carefully.
3. Arrange conductors according to the T568B standard.
4. Insert conductors into RJ45 connectors.
5. Crimp both ends correctly.
6. Test the cable using a cable tester.

7. Demonstrate functionality to the assessor.

ELEMENT COVERED: 3 TEST COMPUTER NETWORK

TASK 2: CAPTURE ICMP PACKETS USING WIRESHARK (15 MARKS)

Using a computer connected to the network, capture ICMP traffic using Wireshark.

Instructions

1. Launch Wireshark.
2. Select the active network interface.
3. Start packet capture.
4. Open Command Prompt.
5. Execute the command:

Ping 8.8.8.8

6. Stop the packet capture.
7. Apply the ICMP display filter:

Icmp

8. Identify at least one ICMP Echo Request and one ICMP Echo Reply packet.
9. Take a screenshot showing the filtered ICMP packets.
10. Save the capture file as instructed by the assessor.

TASK 3: CAPTURE AND FILTER DNS PACKETS USING WIRESHARK

Capture DNS traffic and filter the packets using Wireshark.

Instructions

1. Start a new packet capture in Wireshark.
2. Open a web browser.
3. Visit any website (e.g., *www.google.com).*
4. Stop packet capture.
5. Apply the DNS display filter:

Dns

6. Identify:
   - DNS Query packet
   - DNS Response packet
7. Take a screenshot showing the DNS packets.
8. Save the capture file as instructed by the assessor.